Chapter 1 Hilbert Space

Chapter 2

2.1 Topological Vector Space

Definition 2.1.0.1 (Topological Vector Space) Suppose τ be a topological space on vector space V such that:

- Every point $x \in X$, $\{x\}$ is closed;
- The vector space operations are continuous with respect to τ. $(E, +, \cdot)$.

Chapter 3 Semi-Norms

3.1 Semi-Norms and Locally Convex Linear Topological Spcaes

3.2 Space $\mathcal{D}(\Omega)$

Definition 3.2.0.1 Take the following notation:

- $C_0^\infty(\Omega)$ with a $\mathbb{R}$-module structure;
- $D_k(\Omega) := \{f \in C_0^\infty | supp(f) \subseteq K\}$.

Define the seminorm on $D_K(\Omega)$ by:

$$p_{k,m}(f) := \sup_{|s| \leq m, x \in K} |D^s f(x)|, m < \infty$$